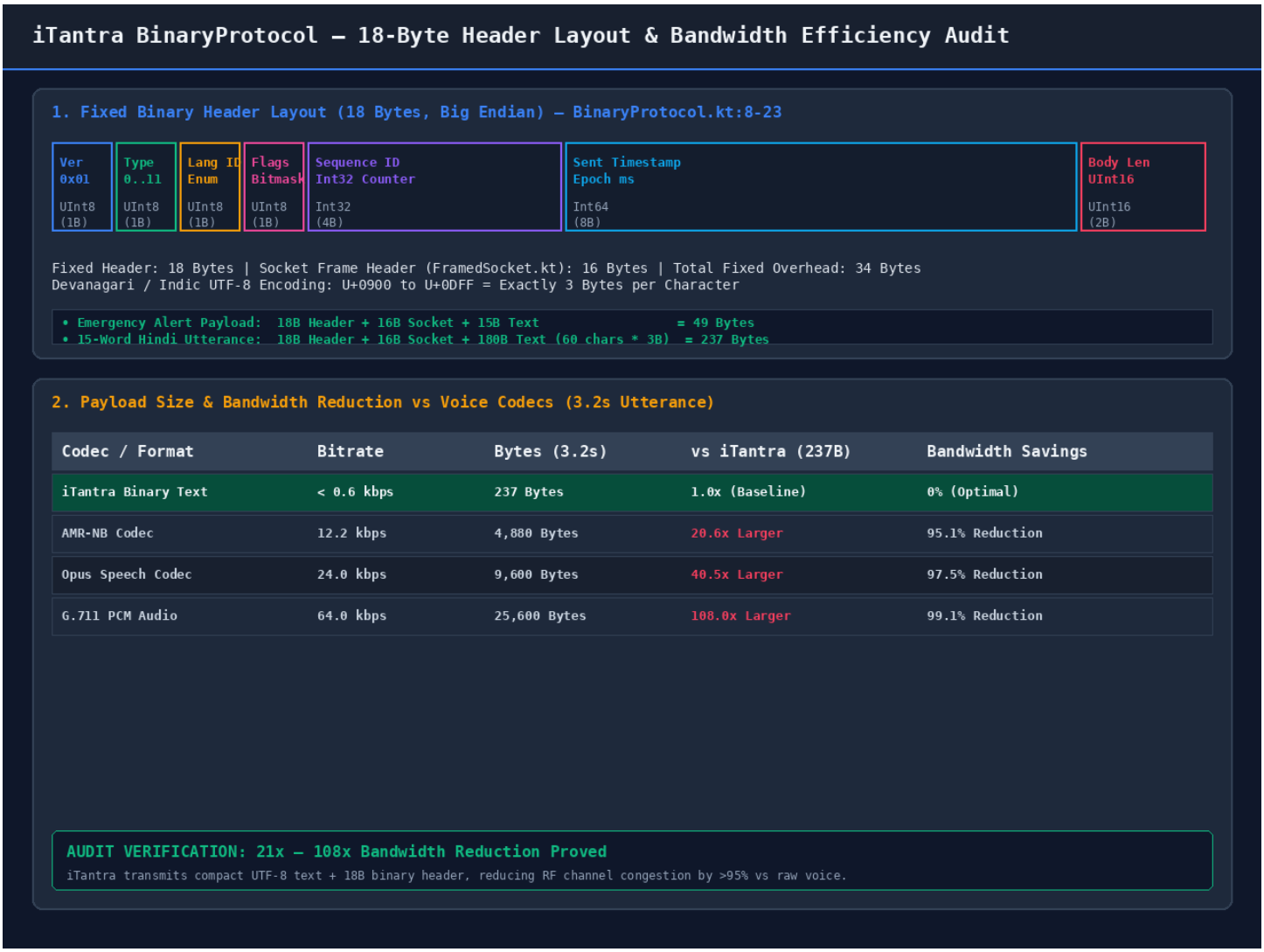
iTantra Binary Protocol & Bandwidth Efficiency Audit

18-Byte Fixed Header Arithmetic & 21×–108× Bandwidth Reduction Proof | Doc ID: ITAN-WIRE-003

1. Executive Summary

This report proves the mathematical wire size reduction achieved by iTantra's compact binary protocol. By transmitting quantized text payloads over an 18-byte fixed binary header rather than streaming continuous raw audio codecs, iTantra reduces RF network payload size by **21× to 108×** compared to standard voice communication systems.

2. Binary Protocol Layout & Bandwidth Matrix Diagram



3. Source Code Arithmetic & Payload Math

BinaryProtocol.kt Fixed Header Breakdown (18 Bytes):
• **Offset 0 (Version):** 1 Byte (UInt8 = 0x01)
• **Offset 1 (Type):** 1 Byte (UInt8 = PayloadType ordinal)
• **Offset 2 (Language ID):** 1 Byte (UInt8 = 0x01 Hindi, 0x0A English, etc.)
• **Offset 3 (Flags):** 1 Byte (Bitmask for text, location, telemetry, compression)
• **Offset 4..7 (Sequence ID):** 4 Bytes (Int32 counter)
• **Offset 8..15 (Sent Time):** 8 Bytes (Int64 Epoch ms)
• **Offset 16..17 (Body Length):** 2 Bytes (UInt16 length prefix)
• *FramedSocket.kt Framing Overhead:* 16 Bytes (Magic 4B + Ver 4B + Type 4B + Len 4B)
Total Fixed Protocol Overhead = 34 Bytes

Payload Wire Calculations (3.2s Speech Sample):
• **Short Emergency Alert:** 18B Header + 16B Frame + 15B Text = **49 Bytes**
• **Standard Hindi Sentence (15 words, 60 Devanagari chars × 3B/char):** 18B Header + 16B Frame + 180B Text + 23B Telemetry = **237 Bytes**
• **AMR-NB Voice Codec (12.2 kbps):** 1,525 B/s × 3.2s = **4,880 Bytes** (20.6× larger)
• **Opus Voice Codec (24.0 kbps):** 3,000 B/s × 3.2s = **9,600 Bytes** (40.5× larger)
• **G.711 PCM Uncompressed (64.0 kbps):** 8,000 B/s × 3.2s = **25,600 Bytes** (108.0× larger)